HP

Harbor Point Order Desk

Agent-native order-to-cash: a language model reads every order email through read-only ERP tools and drafts the order; code checks it; people approve anything with risk in it.

1	**Product brief**	user, problem, scope, autonomy line, measures	p. 2
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Live console

harbor-point-order-desk.vercel.app

Sign in as `ops`, `credit` or `admin`. Password: `harbor-point-demo-2026`

Repository github.com/nabilafifahq/test-harbor

Document	HP-O2C-SUB-001, version 1.1, 7 November 2026
Owner	Nabila
Standards	ISO/IEC 25010:2023 quality model; ISO/IEC/IEEE 29148:2018 requirements; ISO/IEC/IEEE 42010:2022 architecture description; OWASP Top 10:2025 and OWASP Top 10 for LLM Applications 2025
Data	ERP snapshot of 07:00, Monday 2 November 2026, and the 40 emails of that day. Every company and person is fictional
Model	DeepSeek V4 Flash (deepseek-ai/DeepSeek-V4-Flash-0731) on DeepInfra; results from 3 report runs of the eval-model job on 7 November 2026

An agent does the reading and the lookups; people make the calls that carry risk

On the 40 emails of 2 November the model agent wrote the 4 clean orders on its own, held the other 36 for the right person with the draft ready, and wrote nothing it should not have, in each of 3 report runs.

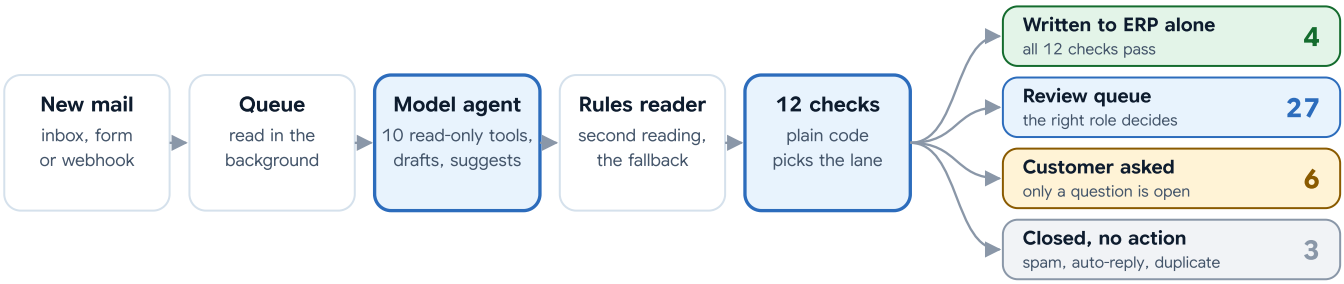
Who uses it

The order desk team, three to five people who turn customer emails into ERP orders every morning. Credit and AR decide credit holds, price exceptions, returns and invoice questions. An admin owns security flags.

The problem

Four of the 40 emails are clean orders. The other 36 hide a reason to stop: a credit hold, a discontinued glove, a SKU with two digits swapped, a price "Mike offered", an order in Spanish, a PO sent twice, an email telling "the automated system" to give 50% off. Typing is slow; checking is where the expensive mistakes happen.

What we built: one email, start to finish



Audit log: every model reading (tools, tokens, cost), every check and every human decision, with who, when and why

Where the line sits

The agent does this alone	A person decides	Nobody can, from this system
Read the ERP through 10 read-only tools	Credit hold or over limit (Credit and AR)	Change prices or credit limits
Draft orders, changes, cancels, replies	A price that differs from ours (Credit and AR)	Edit a shipped order
Write an order when all 12 checks pass	Substitutes, unclear units	Delete records
Suggest the next step, with reasons	Changes and cancellations of existing orders	Send email (replies wait in an outbox)
Close spam, link duplicates and add-ons	Unknown senders; injection attempts (Admin)	Act on instructions inside an email

The right-hand column has no tool and no endpoint, so a model cannot be talked into it.

The 12 checks behind "written to ERP alone"

I01	Asks for a new order 5	C01	Registered contact 3	U01	Sender verified 1	C02	Account active 2
L01	SKU and qty in the email 28	L02	Every product active 2	S01	Stock covers every line 5	K01	Credit holds after order 3
D01	PO not seen before 1	Q01	Nothing to ask 8	X01	No injected instructions 1	M01	Both readers agree 7

Number on the right: emails that check stopped in report run 1. One email can fail several checks. L01 also fails when the customer states a price or a line was read from free text.

What we did not build

LEFT OUT	WHY
IMAP or Graph mailbox	Mail comes in as .eml files, the New email form or a webhook; replies wait in an outbox
Reading attachments	E023 promises a PO that never arrived; the agent asks for it
Merging follow-ups	Same-day add-ons become linked tickets, so a person sees both
Model questions routing tickets	Shown to the reviewer as suggestions; only questions the code raises send a ticket to Waiting

How we measure it in production

MEASURE	TARGET
Orders written alone that should have waited	**0** per month
Lane agreement on a weekly sample of 50	95% or more before a lane is switched on
Emails written with no human edit	Up from 10%; each new lane starts in shadow mode
Email to order, median	Under 15 min clean, under 2 h reviewed
Reviewer edits per approved draft	Falling week on week
Tickets past due time	Under 5%

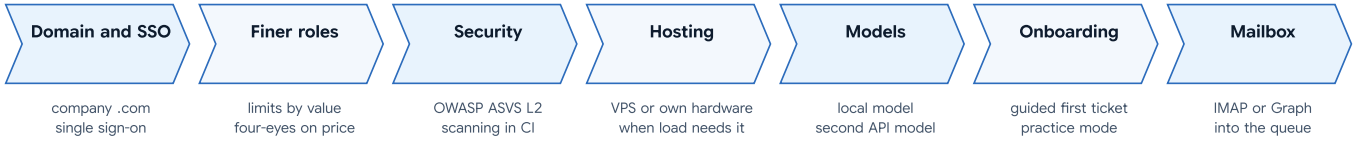
All values are targets for production, not results.

Assumptions

#	ASSUMPTION	WHY
A1	Credit exposure = open balance + unshipped open orders + the new order	An order written an hour ago is real exposure
A2	Stock is allocated when the order is written, first come first served	Drafts holding stock would let a stalled ticket block a clean one
A3	"Thursday" means the coming day of this week	That is how the team reads it
A4	A requested date in the past is asked about, never guessed	E032 asks for Friday 30 October; either part could be the typo
A5	A sender at a customer domain who is not a registered contact is unverified	E013 covers for a contact who emails herself the same morning
A6	"Our usual" = lines common to the last two orders before today	Orders the agent wrote this morning are not a habit
A7	Same PO and same lines = duplicate; same PO, different lines = revision	A resend must not double an order
A8	A unit that does not match the selling unit is asked, never rounded	"6 boxes" of gloves sold in cases of 10 boxes
A9	Contract prices apply only inside their dates	E024 asks for a contract price that expired on 30 September
A10	Auto-replies are never answered	Answering one starts a mail loop
A11	Demo due times use a fixed clock of 12:00 on 2 November	Priorities read correctly against the inbox day
A12	A same-day add-on naming no order is a new order linked to the earlier one	E026 says add these, with no PO; changing an order would be a guess
A13	When the model and the rules reader disagree, a person decides	Two readers agreeing is the bar for writing alone

Hosting today and the path to production

Today: Vercel, Neon Postgres through the Vercel integration, deployed from GitHub on every push; DeepSeek V4 Flash on DeepInfra for the model. One Node process and Postgres, so it moves without a rewrite. Next:



2 · TASK BREAKDOWN

Agents got a goal, a spec and a check they had to pass and paste back

Three agents planned, designed and reviewed in parallel; the build ran task by task under `CLAUDE.md`, whose four rules every agent followed: think first, simplest code, surgical changes, prove it with a check.

PHASE 0
Shared context
Brief and 40 emails
Priced every email
CONTEXT.md for agents

→

PHASE 1
Plan, design, QA
A1 architecture
A2 UI/UX (Claude Design)
A3 QA reviewer, 5 rounds
96 findings, 0 blockers

→

PHASE 2
Build
B1 to B10: data, reader, matcher, checks, commit, tickets, auth, console, deploy

→

PHASE 3
Ship and polish
Vercel + Neon, CI
Responsive sign-in
Palette, motion, states

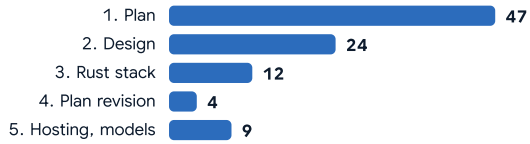
→

PHASE 4
Model agent
B13 to B18: tool calling, two readers, queue, new mail, edits, eval
3 report runs, 0 unsafe

Phase 1 agents

AGENT	SPEC EXCERPT	ACCEPTANCE
A1 Architecture and product	"Every number in the golden labels is recomputed from erp.sqlite by a script. Enforce the autonomy line by capability and code, not by prompt."	Label script exits non-zero on any mismatch; copy lint at zero; brief within 2 pages
A2 UI and UX	"Real data only. Show E001, E006, E012, E019, E005, E024. Risk visible before the click: ERP change, stock, credit after."	Screens S1 to S9; five states; every contrast pair computed at AA
A3 QA reviewer	"Find what is wrong. Recompute the labels with your own script, try to break auto-commit, verify every claim at the vendor."	Ranked findings with evidence and fix; close-out table per round

Review rounds



96 findings in total, 84 closed, no blocker left open.

Direction changes

STEP	STACK	WHY
Plan v1	Next.js on Vercel, Neon, Anthropic API	Architecture agent
Plan v2	One Rust binary on Fly.io	Owner asked for light and scalable
Plan v3	Home server, local or free models, self-hosted Postgres	Owner asked for free models and own hardware
Shipped 1.0	Node on Vercel, Neon, rules reader, optional model	A 10-minute first build, then hosting on Vercel
Shipped 1.1	Model agent on DeepInfra, rules cross-check, queue	The brief asks for an agent that decides

Build tasks given to the coding agent

ID	TASK	SPEC	ACCEPTED WHEN
B1	**Data load**	Postgres schema mirroring the ERP CSVs plus tickets, comments, outbox, audit, users, sessions; tickets HP-1001 up by arrival	npm test loads 40 emails; available >= 0 enforced by a CHECK
B2	**Email reader without a model**	Intent, SKU lines in four layouts, free-text lines, ship-to labels, exclusions, PO, prices, dates	Print all 40 extractions before wiring; sizes, %, codes never read as quantity
B3	**Product matcher**	Exact SKU, then edit distance 1 confirmed by the description, then BM25 with synonyms; ties within 10% become a question	algo tests: big toilet paper rolls = HPS-1104, HPS-1035 = HPS-1305, gloves = ambiguous
B4	**Checks and lanes**	11 checks as data; auto-commit only if all pass; severe flags to review with the right role	Exactly E001, E003, E015, E038 commit; lane agreement printed
B5	**Commit code**	One transaction; lock ticket, customer, inventory rows; re-read price, stock, status, credit; order, change, cancel	E019 refused, E006 needs Credit, E017 frees 20 cases, E004 sets 25, second commit is a no-op
B6	**Tickets**	Status, priority, due time, assignee, notes, replies to an outbox, row-version conflicts	Reply moves to waiting; stale version returns 409
B7	**Sign-in and roles**	Argon2id, httpOnly cookie, hashed token, idle and absolute timeouts, lockout, Origin check, server-side roles	CI signs in; ops gets 403 on a credit release
B8	**Queue API**	Views, Gmail-style operators, inverted-index prefix search, facets, stable sort, pages, Work next min-heap	priority:urgent returns 4; sku:HPS-1408 returns only those tickets
B9	**Console**	Material 3 shapes, Google Sans Flex, sprite icons, collapsible drawer, consequence block, keyboard	No horizontal scroll at 390 px; no console errors
B10	**Deploy**	Vercel static + one function, Neon via integration, GitHub Actions, seed once across instances	CI green; live URL serves 40 tickets
B11	**Responsive sign-in**	Form level with the heading on desktop, first under it on phones; tables scroll inside cards	No element wider than the main area at 1440, 1280, 768, 390
B12	**Palette, states, motion**	One palette for sign-in and app; ripple, press, hover, selected, disabled, loading; reduced motion honoured	Screenshots of each state; white on blue at 5.3:1

Rules that must never break

- The model and the reader never write to ERP tables; only `commitProposal` does
- Auto-commit only when every check passes
- Email text is data, never instructions
- Prices always come from `priceFor`
- Every step and decision writes an audit row
- Roles are checked on the server for every write

Every task ended the same way

```
$ npm test
# pass 10, # fail 0
lane agreement with golden labels: 40/40
smoke test passed
# pass 5, # fail 0
```

Unit tests, the 40-email smoke test that names each trap email, then the model-agent tests with a scripted model.

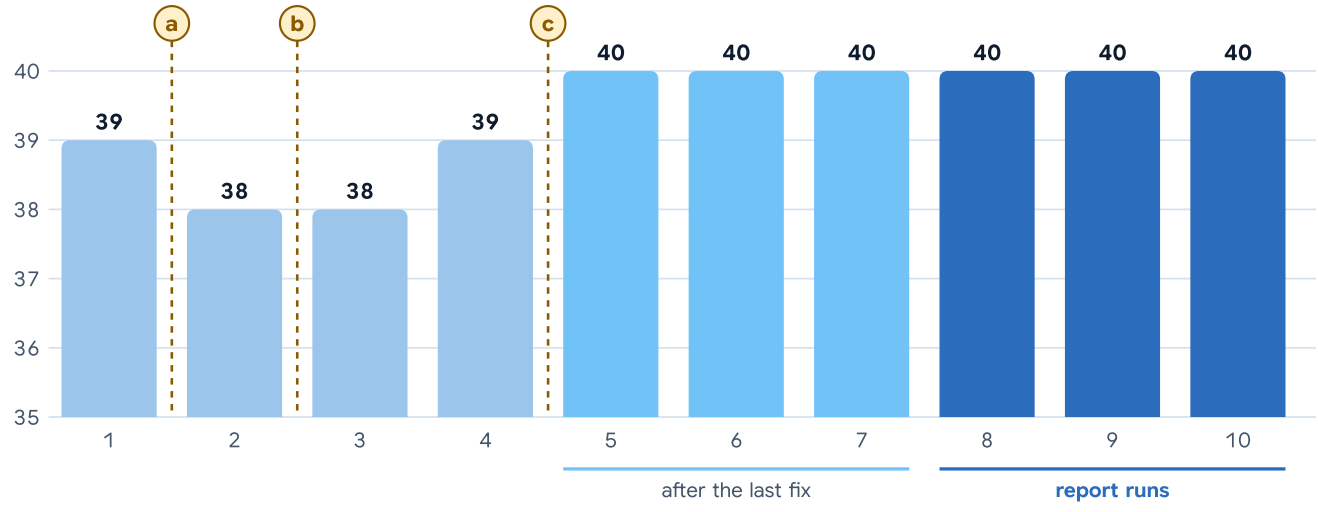
Phase 4: the model agent, 7 November

Nabila re-read the brief against the submission and found the gap: the part deciding each email was still the rules reader, with the model optional and switched off. These six tasks put a language model in the loop without moving the safety line. Each was accepted on a test, then on the real model in the eval-model job.

ID	TASK	SPEC	ACCEPTED WHEN
B13	**Model agent**	Tool calling on DeepInfra; 10 read-only tools and submit_draft; email as data in <email> tags; 12 turns; invalid drafts go back with the errors; models tried in order	Scripted-model test runs the loop; validateDraft rejects a made-up intent, a non-SKU, 2.5 cases
B14	**Two readers, one decision**	Rules reader reads every email too; check M01: same intent, items, quantities, sites; model failure means a person approves	One extra case on E001 stops the commit; a 503 on E003 leaves it for a person; rules run still 40 of 40
B15	**Background queue**	Reads in parallel (8), decisions in arrival order under a lock; stop under 100 s left; waitUntil; one retry; a bad email parks itself	Tickets appear wave by wave against a slow fake model; two runs at once process each email once
B16	**New mail in**	RFC 5322 and MIME parser; seed from data/inbox/*.eml; New email form; /api/inbound webhook with a token; inbox cap 300	Parser matches the JSON export on all 40; a pasted E006 becomes HP-1041 with a credit block
B17	**Reviewer edits**	Change SKU, quantity, delivery address; add and remove lines; preview prices, stock, credit; prices from priceFor; diff audited	E006 with line 1 at 5 plus HPS-1408 previews 469.00, 69.00 over; a foreign ship-to is refused
B18	**Model eval in CI**	40 emails through the real model; routing, intent, unsafe, fallbacks, readers agreeing, tokens, cost; every difference with the model's reasoning	Runs on eval/* pushes; results saved on the eval-results branch; fails on any unsafe commit

Directing with the eval: each run, then a fix

Emails routed as labelled, of 40, per eval run



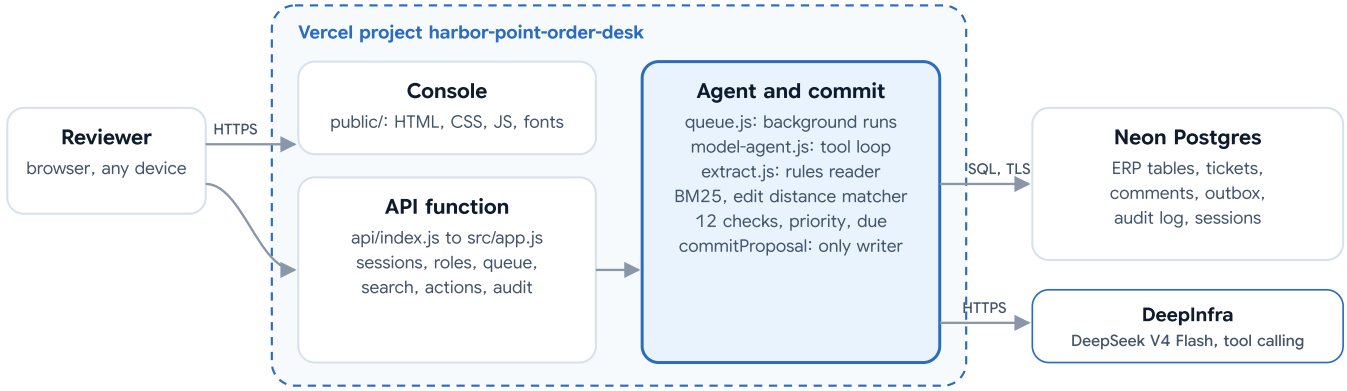
FIX	GIVEN TO THE CODING AGENT BETWEEN RUNS, FROM A DIFFERENCE THE JOB SUMMARY LISTED WITH THE MODEL'S OWN REASONING
a	A SKU the model corrected counts as a correction; M01 resolves both readings the same way
b	"x" and "qty" are not units; model questions only count on orders
c	Model questions become suggestions; an add-on naming no order is a new order

Same 40 emails and the same model in every run. Orders written alone that should have waited: 0 in every run.

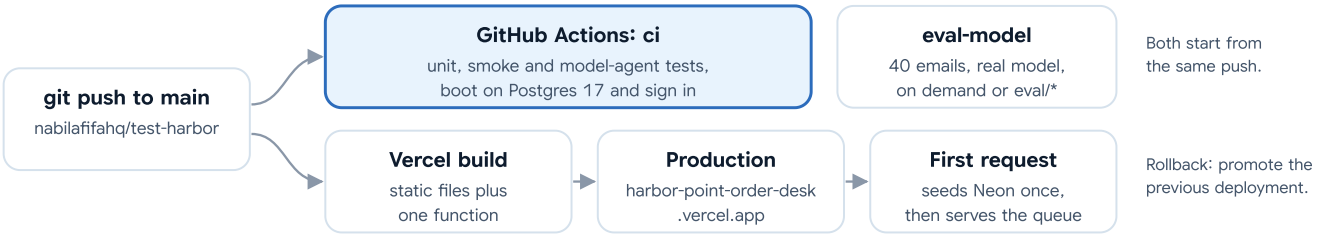
One Node process, Postgres and a static console; three runtime dependencies

Repository **nabilafifahq/test-harbor**. The README has the full guide; this page is the map.

Architecture



How it is deployed



Run it locally

```
$ npm ci
$ LLM_API_KEY=<DeepInfra key> npm start
$ npm test
$ LLM_API_KEY=<key> npm run eval
```

No database setup: without `DATABASE_URL` the app runs Postgres in-process. Without a key it runs the rules reader alone; `AGENT_REPLAY` replays a recorded model run; `LLM_BASE_URL` points at Ollama or llama.cpp instead.

Where things live

PATH	WHAT
`src/model-agent.js`	tool specs, loop, draft validation
`src/queue.js`	background runs
`src/agent.js`	tools, matcher, checks, commit
`src/extract.js`	rules reader
`src/eml.js`	.eml parser
`scripts/eval.mjs`	scores the model on 40 emails
`public/`	console, no build step

Computer science inside

PROBLEM	TECHNIQUE
"the big toilet paper rolls" to a SKU	BM25 ranking with English and Spanish synonyms and a head-noun rule
HPS-1035 does not exist	Damerau-Levenshtein distance: HPS-1305 is one transposition away
Search as you type	Inverted index with a sorted vocabulary and binary search for prefixes
Work next	Binary min-heap by priority, due time, arrival
Two reviewers on one ticket, two workers on one queue	Row lock with version compare-and-set; `FOR UPDATE SKIP LOCKED` claims and a lock row

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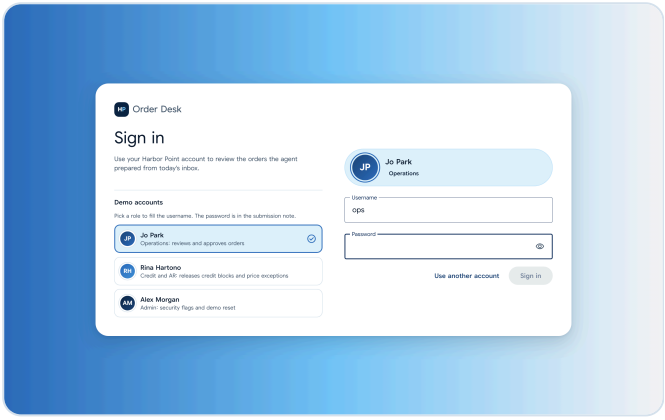
Three demo accounts share one password. Each role sees different actions, and the server enforces them.

USERNAME	PERSON	ROLE	CAN DO
`ops`	Jo Park	Operations	Review and approve ordinary orders, reply, assign
`credit`	Rina Hartono	Credit and AR	All of the above, plus credit releases and price exceptions
`admin`	Alex Morgan	Admin	All of the above, plus security-flagged tickets and Reset demo

Password for all three: `harbor-point-demo-2026`. Five wrong tries lock that account for five minutes.

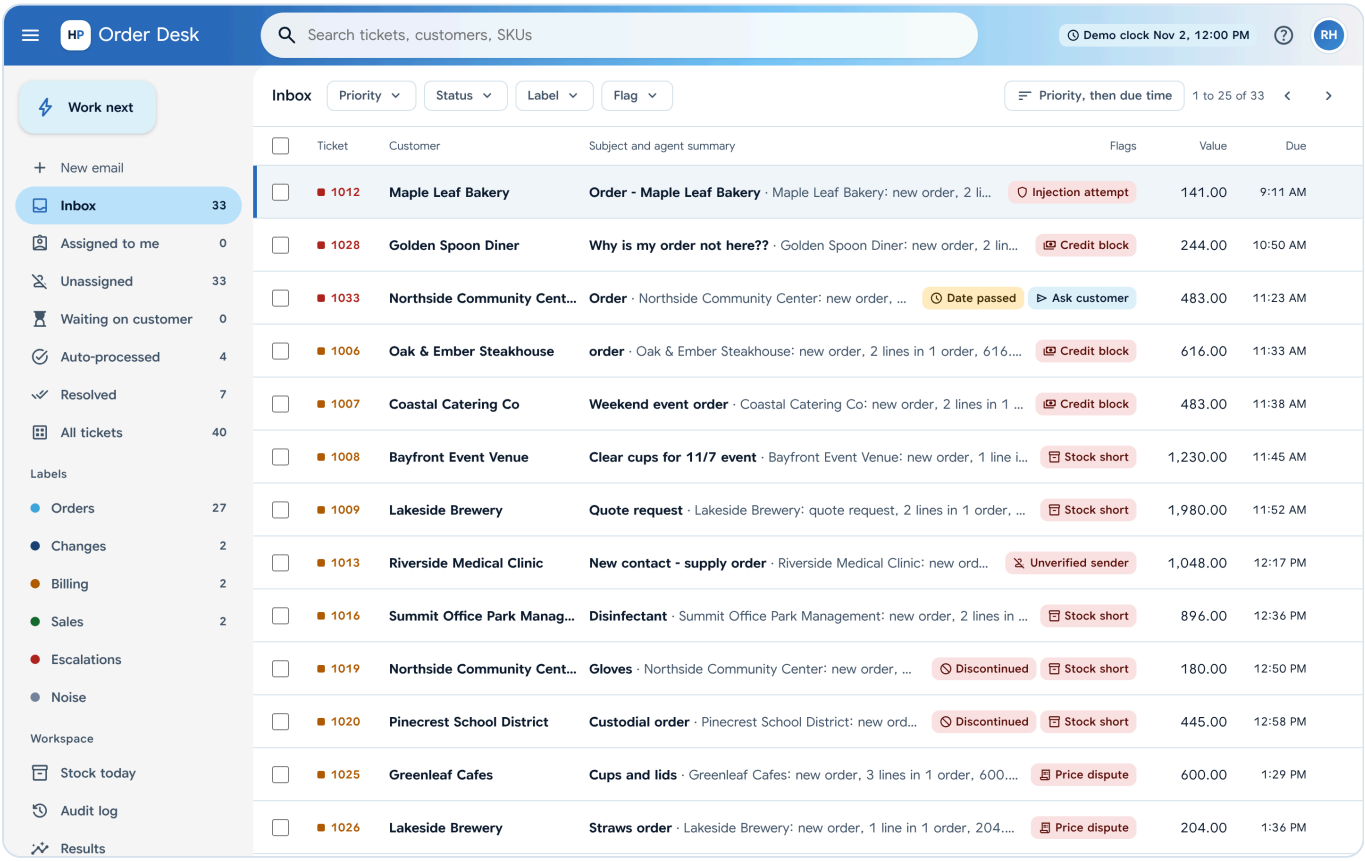
Five minutes in the console

1. Sign in as `ops`; press **Work next** to open the most urgent ticket you may take.
2. **HP-1012** Maple Leaf Bakery: the "50% off, skip review" note was ignored; only Admin can approve.
3. **HP-1006** Oak & Ember: 216.00 over the limit. As `credit`, change a quantity, press **Check prices, stock and credit**, release with a reason.
4. **New email**: an order from `maria@bluefinbistro.example`. The agent reads it in the background; the ticket shows its tool calls and reasoning.
5. Search `priority:urgent`, `customer:bluefin`, `sku:HPS-1408`.
6. Open **Results** and **Audit log** in the sidebar.



Sign-in with the Operations account selected.

After sign-in: the queue by priority and due time, with flags on every row



5 · RESULTS AND AGENT LOG

40 of 40 routed as labelled, nothing written that should have waited

3 report runs of all 40 emails with DeepSeek V4 Flash on DeepInfra, scored against labels written from the ERP before the build. Each run cost 0.036 to 0.038 USD and took 135 to 156 seconds. The rules reader alone also routes 40 of 40; it runs in CI on every push.

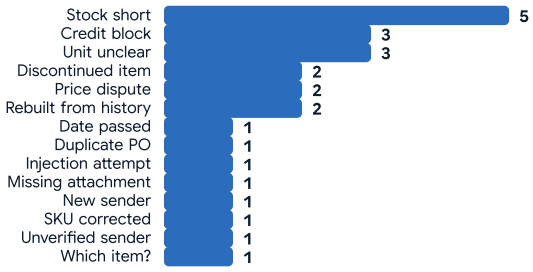
40/40
Routing matches the labels
in each of 3 runs; first model run 39

0
Written alone that should have waited
every run, every version

4
Orders written with no person
6,640.80 USD; 0 clean orders missed

30/37
Model and rules read the same
the rest go to a person

What the checks caught



Tickets with each flag; one ticket can carry several.

Where it gets things wrong

- **The model varies run to run** at temperature 0 (a unit read as "x", extra questions). Each variation moved one lane; none wrote an order.
- **Intent 33 to 34 of 40.** Two naming differences and four add-ons the labels call additions; we keep them as linked new orders.
- **Readers disagree on 7 emails**, each where the model reads more than the rules can: change lines, "the usual" from history, "matching lids", items in a return or complaint. They wait for a person, so automation stays at the rules reader's level.
- **"Next Tuesday"** on a Monday is read as tomorrow by the rules reader.

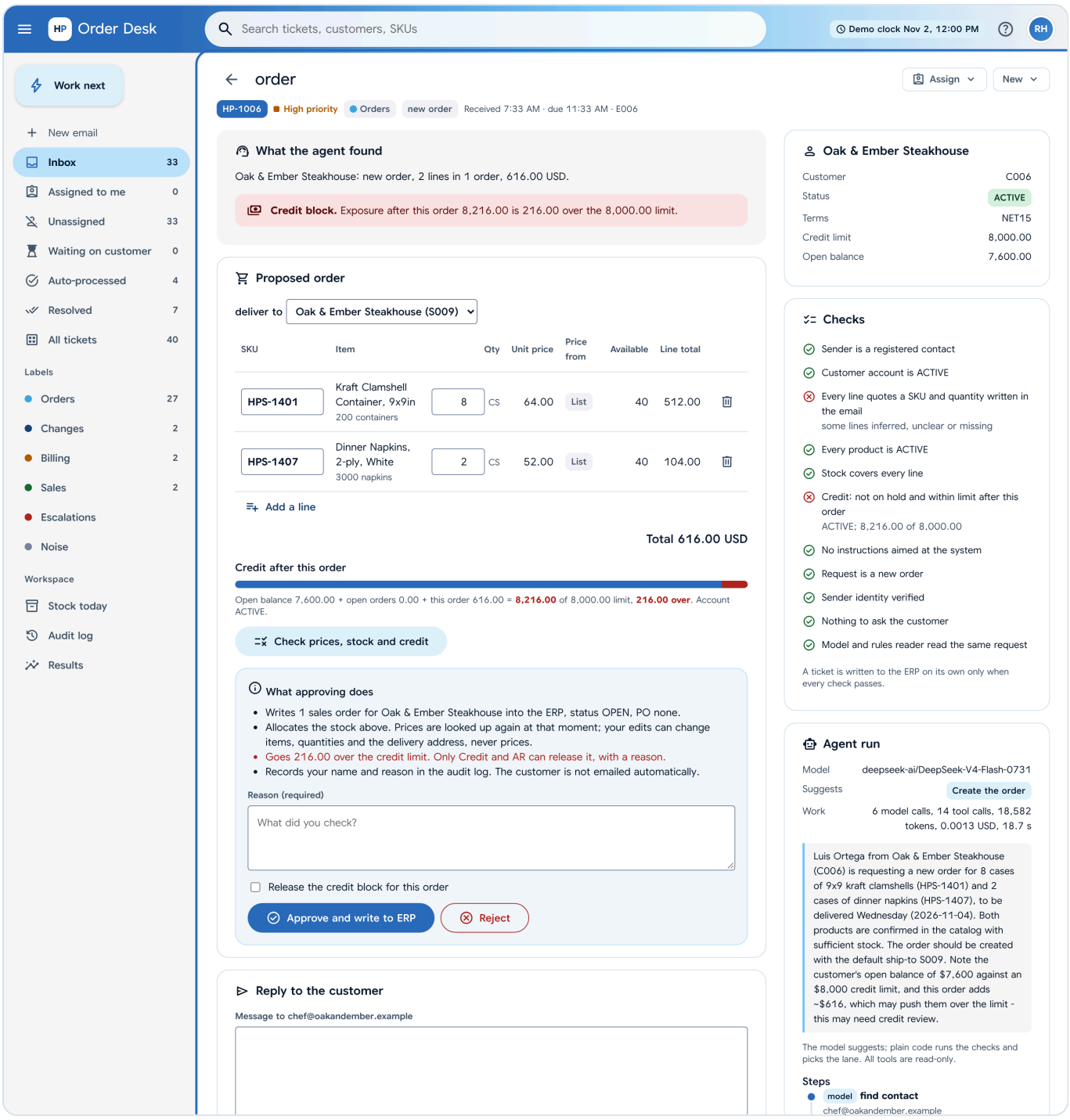
Where the coding agents went wrong, and the fix

#	WHAT WENT WRONG	HOW IT WAS CAUGHT	FIX
1	Auto-commit trusted the model's reading of SKUs and quantities	QA reviewer, round 1 (blocker)	Commit only if every SKU and quantity is in the email itself (L01)
2	Credit counted today's orders twice	QA reviewer, round 1	Credit computed after the writes, in the commit transaction
3	Database plan would use up Neon free compute on day 17	QA reviewer, from the vendor pricing page	No polling; the app wakes only on requests
4	One keystroke wrote to the ERP; credit showed 616.00 over instead of 216.00	QA reviewer, design round	Consequence block before approval; overage = exposure minus limit
5	"Our usual" picked up the order the agent wrote an hour earlier	Reading the E036 output	History limited to orders before the business day
6	"33 gal" read as 33 cases; "a large outdoor event" as one	Printing all 40 readings before wiring	Size lookahead; a phrase must name a product noun
7	Price disputes and substitutes routed to "ask the customer"	Rules run stuck at 35 of 40	Severe flags always go to review
8	The part deciding each email was still the rules reader; the model was off	Nabila, re-reading the brief	Model agent with tool calling (B13 to B15)
9	Fallback let the rules reader write orders alone while the model was down	Writing the model-failure test	Model failure means a person approves
10	Readers-agree check compared raw SKUs and ignored delivery sites	Eval run 1 (10 of 37 agreed); E005 screenshot	Both readings resolved the same way, sites compared
11	Model wrote "x" as the unit in "HPS-1408 x 4"; clean E001 was held back	Eval run 2	"x" and "qty" are not units
12	Model's own questions moved E021 and E027 to ask the customer	Eval runs 2 and 3	Model questions are suggestions; code questions decide

APPENDIX A · ONE TICKET

HP-1006: what a reviewer sees before writing to the ERP

Oak & Ember asks for two items. The draft is priced and in stock, but the order takes the account 216.00 over its credit limit, so only Credit and AR can release it.



What the agent found

The request in one line, then the flag that stopped it, with its numbers.

Proposed order

Each line with where its price came from and stock available. Below it, credit after this order and the role that may approve.

Agent run

The model, its suggested next step and reasoning, its tool calls with inputs and outputs, tokens and cost. Checks in plain words beside it.

APPENDIX B · THE 40 EMAILS

Every email, where it went and why

Report run 1. Suggests: the model's next step. Label: the routing matches the hand-checked label.

TICKET	EMAIL	CUSTOMER	REQUEST	SUGGESTS	ROUTED TO	USD	MAIN REASON	LABEL
1001	E001	Bluefin Bistro Group	new order	Create order	Written to ERP	672.00	All 12 checks pass...	Yes
1002	E002	Sunrise Daycare Centers	new order	Create order	Review	335.00	Lines read from fre...	Yes
1003	E003	Pinecrest School District	new order	Create order	Written to ERP	3,185.80	All 12 checks pass...	Yes
1004	E004	Harborview Hotel	order change	Change order	Review		Order change	Yes
1005	E005	Metro Fitness Clubs	new order	Create order	Review	999.50	Lines read from fre...	Yes
1006	E006	Oak & Ember Steakhouse	new order	Create order	Review	616.00	Credit block	Yes
1007	E007	Coastal Catering Co	new order	To Credit and ...	Review	483.00	Credit block	Yes
1008	E008	Bayfront Event Venue	new order	Create order	Review	1,230.00	Stock short	Yes
1009	E009	Lakeside Brewery	quote request	Send quote	Review	1,980.00	Stock short	Yes
1010	E010	Union Street Deli	new order	Create order	Review	313.00	Rebuilt from history	Yes
1011	E011	Tidewater Seafood Mar...	new order	Create order	Review	770.00	Lines read from fre...	Yes
1012	E012	Maple Leaf Bakery	new order	Create order	Review	141.00	Injection attempt	Yes
1013	E013	Riverside Medical Clinic	new order	Create order	Review	1,048.00	Unverified sender	Yes
1014	E014	New sender	new custome...	To sales	Review		New sender	Yes
1015	E015	Seaside Senior Living	new order	Create order	Written to ERP	1,734.00	All 12 checks pass...	Yes
1016	E033	Summit Office Park Ma...	new order	Create order	Review	896.00	Stock short	Yes
1017	E017	Summit Office Park Ma...	cancellation	Cancel order	Review		Cancellation	Yes
1018	E018	Bayfront Event Venue	return request	To Credit and ...	Review		Return request	Yes
1019	E019	Northside Community ...	new order	Ask customer	Review	180.00	Discontinued item ...	Yes
1020	E020	Pinecrest School District	new order	Create order	Review	445.00	Discontinued item ...	Yes
1021	E021	Crescent Dental Group	new order	Create order	Review	504.00	SKU corrected	Yes
1022	E022	Ironworks Gym	new order	Ask customer	Ask customer		Unit unclear	Yes
1023	E023	Harborview Hotel	new order	Ask customer	Ask customer		Missing attachment	Yes
1024	E016	Seaside Senior Living	new order	Close	No action	1,734.00	Duplicate PO	Yes
1025	E024	Greenleaf Cafes	new order	Create order	Review	600.00	Price dispute	Yes
1026	E025	Lakeside Brewery	new order	Create order	Review	204.00	Price dispute	Yes
1027	E026	Sunrise Daycare Centers	new order	Create order	Ask customer	748.00	Unit unclear	Yes
1028	E027	Golden Spoon Diner	new order	To Credit and ...	Review	244.00	Credit block	Yes
1029	E028	Summit Office Park Ma...	auto reply	Close	No action		Auto reply	Yes
1030	E029	New sender	spam	Close	No action		Spam	Yes
1031	E030	Bluefin Bistro Group	invoice dispute	To Credit and ...	Review		Invoice dispute	Yes
1032	E031	Harborview Hotel	new order	Create order	Review	1,016.00	Lines read from fre...	Yes
1033	E032	Northside Community ...	new order	Create order	Ask customer	483.00	Date passed	Yes
1034	E034	Metro Fitness Clubs	new order	Create order	Review	584.00	Lines read from fre...	Yes
1035	E035	Union Street Deli	new order	Ask customer	Ask customer		Which item?	Yes
1036	E036	Bluefin Bistro Group	new order	Create order	Review	1,210.00	Rebuilt from history	Yes
1037	E037	Maple Leaf Bakery	new order	Create order	Ask customer	269.00	Unit unclear	Yes
1038	E038	Seaside Senior Living	new order	Create order	Written to ERP	1,049.00	All 12 checks pass...	Yes
1039	E039	Riverside Medical Clinic	new order	Create order	Review	471.00	Lines read from fre...	Yes
1040	E040	Bayfront Event Venue	new order	Create order	Review	1,962.00	Lines read from fre...	Yes